

1. Periodic classification of elements is used to examine

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- (a) periodic trends in physical properties of elements
- (b) periodic trends in chemical properties of elements
- (c) both (a) and (b)
- (d) none of the above

Answer: (c) both (a) and (b)

Explanation: Periodic classification arranges elements so that recurring trends in both physical properties and chemical properties can be studied systematically. Therefore, both types of periodic trends are examined.

2. Which of the following is a set containing man-made elements?

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- (a) Gold, Mercury, Zinc
- (b) Gold, Radium, Palladium
- (c) Curium, Osmium, Uranium
- (d) Technetium, Neptunium, Einsteinium

Answer: (d) Technetium, Neptunium, Einsteinium

Explanation: Technetium, neptunium and einsteinium are commonly treated as man-made/synthetic elements in this context. The other options contain naturally occurring elements such as gold, mercury, zinc, osmium or uranium.

3. * Cl, Br and I form a Dobereiner's triad. If the atomic masses of Cl and I are 35.5 and 127, respectively, the atomic mass of Br is

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- (a) 162.5
- (b) 91.5
- (c) 81.25
- (d) 45.625



Answer: (c) 81.25

Explanation: For a Dobereiner triad, the atomic mass of the middle element is approximately the arithmetic mean of the other two. Thus, mass of Br = $(35.5 + 127)/2 = 81.25$.

4. If two members of a Dobereiner triad are phosphorus and antimony, the third member of this triad is

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- (a) arsenic
- (b) sulphur
- (c) iodine
- (d) calcium

Answer: (a) arsenic

Explanation: Phosphorus (P), arsenic (As) and antimony (Sb) form a Dobereiner-type family and all belong to group 15. Therefore, arsenic is the missing member.

5. * Lothar Meyer plotted physical properties such as atomic volume, melting point and ...X... against atomic weight. Here, X refers to

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- (a) mass
- (b) boiling point
- (c) surface tension
- (d) none of these

Answer: (b) boiling point

Explanation: Lothar Meyer studied periodicity by plotting physical properties, including atomic volume, melting point and boiling point, against atomic weight. Hence X is boiling point.

6. * The most significant contribution towards the development of the periodic table was made by

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- (a) Mendeleev
- (b) Avogadro
- (c) Dalton
- (d) Cavendish

Answer: (a) Mendeleev

Explanation: Mendeleev made the most significant early contribution by formulating a periodic law, arranging known elements systematically and leaving gaps for undiscovered elements whose properties he predicted.

7. Who developed the long form of the periodic table?

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- (a) Lothar Meyer
- (b) Niels Bohr
- (c) Mendeleev
- (d) Moseley

Answer: (b) Niels Bohr

Explanation: The long-form arrangement is associated with Bohr's electronic-configuration-based interpretation of the periodic table. Moseley established atomic number as the basis of the modern periodic law, but the long-form table is credited here to Niels Bohr.

8. Which fact is not valid for Dobereiner's triads?

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- (a) The atomic weight of the middle element is roughly the average of the other two elements.
- (b) The properties of the middle element are roughly the average of the other two elements.
- (c) The elements of triads belong to the same group of the modern periodic table.
- (d) The elements of triads have the same number of valence electrons.



Answer: (b) The properties of the middle element are roughly the average of the other two elements.

Explanation: Dobereiner's numerical rule specifically concerned the atomic mass of the middle element being approximately the average of the other two. The three elements also show similar chemical properties and generally lie in the same modern group; their properties are not defined as a numerical average in the same way.

9. * Select the correct chronological order for the discoveries of the following scientists: Dobereiner, Newlands, Chancourtois and Mendeleev.

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- (a) Chancourtois, Dobereiner, Newlands, Mendeleev
- (b) Dobereiner, Chancourtois, Newlands, Mendeleev
- (c) Dobereiner, Newlands, Chancourtois, Mendeleev
- (d) Chancourtois, Newlands, Dobereiner, Mendeleev

Answer: (b) Dobereiner, Chancourtois, Newlands, Mendeleev

Explanation: The important dates are approximately: Dobereiner (1829), de Chancourtois (1862), Newlands (1864–65) and Mendeleev (1869). This gives the order Dobereiner → Chancourtois → Newlands → Mendeleev.

10. * The molecular formulae of the chlorides of Eka-Aluminium and Eka-Silicon, respectively, are

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- (a) GaCl_3 and SiO_4
- (b) GaCl_3 and AlCl_3
- (c) AlCl_3 and SiCl_4
- (d) GaCl_3 and GeCl_4

Answer: (d) GaCl_3 and GeCl_4

Explanation: Eka-aluminium was later identified as gallium (Ga), which commonly forms GaCl_3 . Eka-silicon was identified as germanium (Ge), which forms GeCl_4 .



11. In Mendeleev's periodic table, which of the following elements, instead of having a lower atomic weight, was placed after an element of higher atomic weight, thereby ignoring the order of increasing atomic weights?

- (a) Iodine
- (b) Antimony
- (c) Bromine
- (d) Molybdenum

Answer: (a) Iodine

Explanation: Mendeleev placed tellurium (atomic mass about 127.6) before iodine (about 126.9) because their chemical properties demanded that order. Thus iodine, despite its slightly lower atomic mass, was placed after the heavier element tellurium.

12. Which of the following is correct about Eka-Aluminium and Eka-Silicon?

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- (a) The oxide of Eka-Aluminium is Ga_2O_3 and the oxide of Eka-Silicon is GeO_2 .
- (b) The oxide of Eka-Aluminium is Al_2O_3 and the oxide of Eka-Silicon is Si_2O_3 .
- (c) The melting point of Eka-Aluminium is lower than the melting point of Eka-Silicon.
- (d) Both (a) and (c)

Answer: (d) Both (a) and (c)

Explanation: Eka-aluminium corresponds to Ga and forms Ga_2O_3 ; eka-silicon corresponds to Ge and forms GeO_2 . Gallium also has a much lower melting point than germanium, so both statements (a) and (c) are correct.

13. Which of the scientists given below discovered that the periodic table should be based on atomic number?

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- (a) Mendeleev
- (b) Newlands
- (c) Moseley
- (d) Lothar Meyer



Answer: (c) Moseley

Explanation: Henry Moseley showed from X-ray spectra that atomic number, rather than atomic mass, is the fundamental basis for periodic classification. This led to the modern periodic law.

14. How many elements are there in the 6th period of the periodic table?

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- (a) 18
- (b) 8
- (c) 30
- (d) 32

Answer: (d) 32

Explanation: The sixth period runs from Cs ($Z = 55$) to Rn ($Z = 86$) and includes the lanthanoids. Therefore it contains 32 elements.

15. The long form of the periodic table consists of

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- (a) seven periods and eight groups
- (b) seven periods and eighteen groups
- (c) eight periods and eighteen groups
- (d) eighteen periods and eight groups

Answer: (b) seven periods and eighteen groups

Explanation: The modern long-form periodic table has 7 horizontal periods and 18 vertical groups.

16. Modern periodic table is based on the atomic number of the elements. The experiment that proved the significance of atomic number was

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- (a) Millikan's oil-drop experiment
- (b) Moseley's work on X-ray spectra
- (c) Bragg's work on X-ray diffraction



(d) Discovery of X-rays by Röntgen

Answer: (b) Moseley's work on X-ray spectra

Explanation: Moseley found a regular relationship between the frequency of characteristic X-rays and atomic number. His results demonstrated that atomic number is the correct basis for arranging elements.

17. * The symbol and IUPAC name for the element with atomic number 120, respectively, are

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- (a) Ubn and unbinilium
- (b) Ubn and unbiunium
- (c) Ubn and unnibium
- (d) Ubn and unnilium

Answer: (a) Ubn and unbinilium

Explanation: Using the systematic temporary IUPAC naming rules: 1 = un, 2 = bi and 0 = nil. Therefore 120 gives un-bi-nil-ium, written unbinilium, with symbol Ubn.

18. What is the IUPAC name of the element with atomic number 114?

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- (a) Unununnium
- (b) Unnilquadium
- (c) Ununquadium
- (d) Unnilennium

Answer: (c) Ununquadium

Explanation: For atomic number 114, the systematic digit roots are 1 = un, 1 = un and 4 = quad, giving ununquadium (temporary name used in this nomenclature system).

19. * The IUPAC nomenclature of an element with electronic configuration $[Rn] 5f^{14} 6d^1 7s^2$ is

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- (a) Unnilbium



- (b) Unnilunium
- (c) Unnilquadium
- (d) Unniltrium

Answer: (d) Unniltrium

Explanation: The configuration $[Rn] 5f^{14} 6d^1 7s^2$ contains $86 + 14 + 1 + 2 = 103$ electrons, so $Z = 103$. The systematic name for 103 is un-nil-tri-um = unniltrium.

20. * An element with electronic configuration $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6 4d^{10} 5s^2 5p^3$ belongs to which group of the periodic table?

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- (a) 5th
- (b) 15th
- (c) 3rd
- (d) 17th

Answer: (b) 15th

Explanation: The configuration ends in $5s^2 5p^3$. For a p-block element, $ns^2 np^3$ corresponds to five valence electrons and therefore group 15.

